

A RANK-SENSITIVE STRENGTHENING OF THE DETERMINANTAL IRRATIONALITY CRITERION FOR APÉRY LIMITS

ABSTRACT. Let $A(x) = \sum u_n x^n$ and $V(x) = \sum v_n x^n$ be annihilated by a common differential operator L of order $k \geq 2$, irreducible over $\mathbb{Q}(x)$, with $u_n \in \mathbb{Z}$ and $c[1, \dots, n]^\sigma v_n \in \mathbb{Z}$. Suppose that for some real η the linear-form generating function $\eta A - V \not\equiv 0$ extends holomorphically to the slit plane $\mathbb{C} \setminus [x_2, \infty)$. We observe that the arithmetic holonomy bound of Calegari, Dimitrov and Tang, applied to the tuple $\{1, P, \theta P, \dots, \theta^{k-1} P\}$ with $\theta = x \frac{d}{dx}$, yields: if

$$\log(4x_2) > \sigma \frac{k+2}{k+1},$$

then $\eta \notin \mathbb{Q}$. The case $k = 1$ -analogue $\log(4x_2) > \frac{3}{2}\sigma$ is Zudilin's determinantal criterion, and the classical Apéry-type criterion requires $\log x_2 > \sigma$; the present inequality is strictly weaker than both for every $k \geq 2$, and opens the window $e^{\sigma(k+2)/(k+1)}/4 < x_2 \leq e^\sigma$ in which no previous criterion applies. We tabulate the resulting specification curve for linear forms in 1 and $\zeta(5)$, and we report numerical experiments indicating that the denominators realized by the underlying auxiliary constructions are substantially smaller than the theoretical bound, with the savings concentrated at small primes.

1. INTRODUCTION

Let $(u_n)_{n \geq 0} \subset \mathbb{Z}$ and $(v_n)_{n \geq 0} \subset \mathbb{Q}$, and let $\eta \in \mathbb{R}$ be a target number approximated by the linear forms $\ell_n := u_n \eta - v_n$. Write $[1, \dots, n]$ for $\text{lcm}(1, \dots, n)$ and suppose

$$(1) \quad c[1, \dots, n]^\sigma v_n \in \mathbb{Z} \quad (n \geq 0, \text{ with the convention } [1, \dots, 0] := 1)$$

for fixed positive integers c and σ ; the condition at $n = 0$, namely $cv_0 \in \mathbb{Z}$, may always be arranged by enlarging c , since v_0 is a single rational number. Apéry's method [1, 3] proves $\eta \notin \mathbb{Q}$ as soon as $\limsup |\ell_n|^{1/n} =: 1/x_2$ satisfies $x_2 > e^\sigma$: indeed, if $\eta = p/q$ then $cq[1, \dots, n]^\sigma \ell_n$ are integers of absolute value $\rightarrow 0$, hence eventually zero.

Zudilin [4] extended the reach of the method to situations where the *decay* of ℓ_n is insufficient but the generating function

$$L_\eta(x) := \sum_{n \geq 0} \ell_n x^n = \eta A(x) - V(x), \quad A = \sum u_n x^n, \quad V = \sum v_n x^n,$$

extends holomorphically to the slit plane $\Omega := \mathbb{C} \setminus [x_2, \infty)$: under a positivity hypothesis his determinantal criterion proves $\eta \notin \mathbb{Q}$ whenever $\log(4x_2) > \frac{3}{2}\sigma$. The constant $4x_2$ is the conformal radius of Ω at the origin, and the exponent $\frac{3}{2}$ reflects the arithmetic cost of Hankel determinants. The same inequality reappears, with the positivity hypothesis removed, as the $m = 2$ case of the *univalent arithmetic holonomy bound* of Calegari–Dimitrov–Tang [2, Theorem 2.7.1 and §2.7.7].

The purpose of this note is to record the observation that the *order of the differential equation* satisfied by the linear forms is an arithmetic resource: feeding the derivative tower of the (conditional) rational function $P = pA - qV$ into the holonomy bound of [2] lowers the exponent $\frac{3}{2}$ to $\frac{k+2}{k+1}$, where k is the order of the underlying irreducible operator.

Theorem 1.1. *Let $k \geq 2$ and let L be a linear differential operator of order k with coefficients in $\mathbb{Q}(x)$, irreducible over $\mathbb{Q}(x)$. Let $A = \sum u_n x^n$ and $V = \sum v_n x^n$ in $\mathbb{Q}[[x]]$ be annihilated by L , with $u_n \in \mathbb{Z}$ and with (1) for fixed positive integers c, σ . Let $\eta \in \mathbb{R}$ be such that $L_\eta = \eta A - V$ is not identically zero and extends holomorphically to $\Omega = \mathbb{C} \setminus [x_2, \infty)$ for some $x_2 > 0$. If*

$$(2) \quad \log(4x_2) > \sigma \frac{k+2}{k+1},$$

then $\eta \notin \mathbb{Q}$.

For $\sigma = 3$ (the type of Apéry's $\zeta(3)$ approximations) the thresholds on x_2 read:

criterion	condition	threshold on x_2
classical (radius)	$x_2 > e^\sigma$	20.09
determinantal [4]	$4x_2 > e^{3\sigma/2}$	22.51
Theorem 1.1, $k = 2$	$4x_2 > e^{4\sigma/3}$	13.65
Theorem 1.1, $k = 3$	$4x_2 > e^{5\sigma/4}$	10.63
Theorem 1.1, $k \rightarrow \infty$	$4x_2 > e^\sigma$	5.02

Every $k \geq 2$ row opens territory where neither the classical nor the determinantal criterion concludes. We stress at once that for Apéry's actual $\zeta(3)$ data ($x_2 = (1 + \sqrt{2})^4 \approx 33.97$, $k = 3$) the classical criterion already applies; the content of Theorem 1.1 lies entirely in the window $x_2 \in (e^{\sigma(k+2)/(k+1)}/4, e^\sigma]$, i.e. in its potential applications to constructions of sub-classical quality, for instance towards $\zeta(5)$ (§4).

Theorem 1.1 is a direct application of the published holonomy bound of [2]; the proof below consists of the verification, which we have endeavoured to make complete, that the derivative tower satisfies every hypothesis of that bound, together with the elementary computation of the resulting numerology. Given that the consequence strengthens two classical criteria on a well-trodden battlefield, we submit this note explicitly for expert scrutiny.

2. THE INPUT FROM [2]

We quote the univalent arithmetic holonomy bound in the form we use. For an $m \times r$ array $\mathbf{b} = (b_{i,j})$ of nonnegative reals whose columns consist of u_j zeros followed by a constant value b_j , set $\sigma_i := \sum_j b_{i,j}$ (row sums, nondecreasing) and

$$\tau(\mathbf{b}) := \frac{1}{m^2} \sum_{i=1}^m (2i-1) \sigma_i.$$

Theorem 2.1 ([2, Theorem 2.7.1; see also Theorem 7.0.1 and the holonomic relaxation stated there]). *Let $\Omega \subset \mathbb{C}$ be a contractible domain containing 0, of conformal radius $\rho = \rho(\Omega, 0)$. Let $f_1, \dots, f_m \in \mathbb{Q}[[x]]$ be $\mathbb{Q}(x)$ -linearly independent, holonomic, meromorphic on Ω , with*

$$f_i(x) = a_{i,0} + \sum_{n \geq 1} a_{i,n} \frac{x^n}{[1, \dots, b_{i,1}n] \cdots [1, \dots, b_{i,r}n]}, \quad a_{i,n} \in \mathbb{Z},$$

for the array $\mathbf{b}$ as above, and suppose $\rho > e^{\tau(\mathbf{b})}$. Then

$$m \leq \frac{\log \rho}{\log \rho - \tau(\mathbf{b})}.$$

We will also use the classical fact that the Koebe-type map $\varphi(z) = 4x_2 z / (1 + z)^2$ is a conformal equivalence $\mathbb{D} \rightarrow \mathbb{C} \setminus [x_2, \infty)$ with $\varphi(0) = 0$ and $\varphi'(0) = 4x_2$; thus $\rho(\Omega, 0) = 4x_2$ for the slit plane.

Remark 2.2. The vanishing-filtration input behind Theorem 2.1 is, for the tuple used below, available in its oldest form: the $\mathbb{Q}(x)$ -linear span of $\{1, P, \theta P, \dots, \theta^{k-1} P\}$ is closed under d/dx (see Lemma 3.2), so Shidlovsky's lemma [2, Theorem 3.2.8] applies with no ε -loss, and the system is Fuchsian, so Chudnovsky's effective form [2, Theorem 3.2.10, Remark 3.2.12] applies as well.

3. PROOF OF THEOREM 1.1

Assume $\eta = p/q$ with p, q coprime integers, $q \geq 1$. Set

$$P := q L_\eta = pA - qV \in \mathbb{Q}[[x]], \quad \theta := x \frac{d}{dx}.$$

Then $P \neq 0$, $L(P) = 0$, and P is holomorphic on $\Omega = \mathbb{C} \setminus [x_2, \infty)$.

Lemma 3.1 (Independence). *Let L be irreducible over $\mathbb{Q}(x)$ of order $k \geq 2$ and let $0 \neq P \in \mathbb{Q}[[x]]$ satisfy $L(P) = 0$. Then $1, P, \theta P, \dots, \theta^{k-1} P$ are $\mathbb{Q}(x)$ -linearly independent.*

Proof. Since x is invertible in $\mathbb{Q}(x)$, the span of $\{\theta^j P\}_{j < k}$ over $\mathbb{Q}(x)$ equals that of $\{P, P', \dots, P^{(k-1)}\}$; suppose $r_0 + M(P) = 0$ with $r_0 \in \mathbb{Q}(x)$ and M an operator of order $\leq k-1$ over $\mathbb{Q}(x)$, not both zero. In the left-Euclidean ring $\mathbb{Q}(x)\langle \theta \rangle$ the annihilator of P is the left ideal generated by the minimal operator L_P of P , and every annihilator of P is a left multiple of L_P . As L annihilates P and is irreducible, L_P has order 0 or k ; order 0 forces $P = 0$, so L_P has order k .

If $r_0 = 0$ then $M \neq 0$ annihilates P with order $\leq k-1 < k$, a contradiction. If $r_0 \neq 0$ then $M \neq 0$ (else $r_0 = 0$), and from $M(P) = -r_0$ we get $(\partial - \frac{r'_0}{r_0})M(P) = -r'_0 + \frac{r'_0}{r_0}r_0 = 0$. Thus $N := (\partial - \frac{r'_0}{r_0})M$ is a nonzero annihilator of P of order $\leq k$, hence a left multiple of L_P : $N = c L_P$ with $c \in \mathbb{Q}(x)^\times$ by comparing orders, and therefore $L_P = c^{-1}(\partial - \frac{r'_0}{r_0})M$ admits the right-hand factor M of order $k-1$ with $0 < k-1 < k$, contradicting irreducibility (note $\text{ord } M = k-1$ is forced by $\text{ord } N = k$; and $\text{ord } M = 0$ would make $P \in \mathbb{Q}(x)$, i.e. L_P of order $1 < k$). $\square$

Lemma 3.2 (Types and closure). *For $0 \leq j \leq k-1$, the function $cq \theta^j P$ has coefficients $cq n^j (pu_n - qv_n)$, and $[1, \dots, n]^\sigma \cdot cq n^j (pu_n - qv_n) \in \mathbb{Z}$; thus each $\theta^j P$ is, after multiplication by the fixed integer cq , of the arithmetic type of Theorem 2.1 with row $(1, \dots, 1)$ (σ columns, $b_j = 1$). Moreover the $\mathbb{Q}(x)$ -span of $\{1, P, \theta P, \dots, \theta^{k-1} P\}$ is closed under d/dx .*

Proof. The coefficient statement is immediate from (1) and $u_n \in \mathbb{Z}$; multiplication of a coefficient by the integer n^j preserves integrality against the same denominators. For closure: multiplying L by x^k and rewriting in θ (using $x^i \partial^i \in \mathbb{Z}[\theta]$) expresses $\theta^k P$ as a $\mathbb{Q}(x)$ -combination of $P, \dots, \theta^{k-1} P$; and $\frac{d}{dx} \theta^j P = x^{-1} \theta^{j+1} P$ stays in the span, while $\frac{d}{dx} 1 = 0$. $\square$

Lemma 3.3 (Local normalization at no arithmetic cost). *Write $P = \sum_n c_n x^n$, set $h_0 := P - P(0) \cdot 1$ and, for $1 \leq j \leq k-1$,*

$$h_j := \prod_{i=1}^j (\theta - i) P - (-1)^j j! P(0) \cdot 1.$$

Then $\{1, h_0, h_1, \dots, h_{k-1}\}$ is obtained from $\{1, P, \theta P, \dots, \theta^{k-1}P\}$ by an integral, triangular, unit-diagonal (hence unimodular) change of basis; each $cq h_j$ carries the same arithmetic type (row $(1, \dots, 1)$ of length σ) as $cq \theta^j P$, so the array $\mathbf{b}$, its row sums $(0, \sigma, \dots, \sigma)$, and $\tau(\mathbf{b})$ are unchanged; and $\text{ord}_{x=0} h_j \geq j+1$, so the tuple has pairwise distinct vanishing orders at the origin.

Proof. The n -th coefficient of $\prod_{i=1}^j (\theta - i)P$ is $(n-1)(n-2)\cdots(n-j)c_n$; at $n=0$ this is $(-1)^j j! c_0$, and it vanishes for $1 \leq n \leq j$, whence $\text{ord}_{x=0} h_j \geq j+1$ after the constant subtraction. The operator $\prod_{i=1}^j (\theta - i)$ expands monically in $\mathbb{Z}[\theta]$ (integer Stirling coefficients), and $cq P(0) = cq c_0 \in \mathbb{Z}$ by (1); hence the change of basis is integral, triangular, with unit diagonal, and $\mathbb{Q}(x)$ -linear independence persists. For the type: the coefficients of $cq h_j$ are the integer multiples $(n-1)\cdots(n-j) \cdot cq c_n$ of those of $cq P$ (for $n \geq 1$; the constant term is 0), so integrality against the same denominators $[1, \dots, n]^\sigma$ persists verbatim. Likewise, multiplication of any admissible function by a fixed positive integer preserves the type, and no quantity in Theorem 2.1 depends on such constants. $\square$

Remark 3.4. We stress that Theorem 2.1 imposes no hypothesis on the local behaviour of the f_i at the origin: the vanishing-filtration data entering its proof depends only on the $\mathbb{Q}$ -span of $\{x^s f_i\}$, which Lemma 3.3's unimodular change of basis leaves untouched. The lemma is included so that the application can be read with either basis in mind.

Proof of Theorem 1.1. Apply Theorem 2.1 to the $m := k+1$ functions

$$f_1 = 1, \quad f_{j+2} = cq \theta^j P \quad (0 \leq j \leq k-1),$$

on $\Omega = \mathbb{C} \setminus [x_2, \infty)$, with the array $\mathbf{b}$ having first row 0 and the remaining k rows $(1, \dots, 1)$ of length σ . The hypotheses hold: linear independence by Lemma 3.1; the arithmetic type by Lemma 3.2; holonomicity since each f_i lies in the d/dx -stable finite $\mathbb{Q}(x)$ -module of Lemma 3.2; holomorphy on Ω since P is holomorphic there and θ preserves $\mathcal{O}(\Omega)$; and $\rho(\Omega, 0) = 4x_2$ by the Koebe map. The row sums are $(0, \sigma, \dots, \sigma)$, whence

$$\tau(\mathbf{b}) = \frac{\sigma}{m^2} \sum_{i=2}^m (2i-1) = \sigma \frac{m^2-1}{m^2},$$

and (2) gives $\log \rho = \log(4x_2) > \sigma \frac{m+1}{m} > \sigma \frac{m^2-1}{m^2} = \tau(\mathbf{b})$, so the bound applies and yields

$$k+1 = m \leq \frac{\log(4x_2)}{\log(4x_2) - \tau(\mathbf{b})}.$$

Rearranged, $\log(4x_2) \leq \sigma \frac{m+1}{m} = \sigma \frac{k+2}{k+1}$, contradicting (2). Hence no such p/q exists. $\square$

Remark 3.5. For $k=1$ the tuple $\{1, P\}$ requires $P \notin \mathbb{Q}(x)$, which an irreducible order-1 operator does not guarantee; restoring Zudilin's threshold $\frac{3}{2}\sigma$ in that case requires his positivity hypotheses. For $k \geq 2$, irreducibility replaces positivity altogether.

Remark 3.6. Using the bivalent map $8x_2(z+z^3)/(1+z)^4$ of [2, §2.11] together with the evaluation [2, Lemma 2.11.7] of its Bost–Charles integral $(\log 8 + 4G/\pi, G \text{ the Catalan constant})$ in the multivalent bound [2, (2.5.4)] improves the thresholds further for large k ; we have preferred the clean univalent statement.

4. THE WINDOW, AND A SPECIFICATION CURVE FOR $\zeta(5)$

Theorem 1.1 concludes in the region

$$\frac{e^{\sigma(k+2)/(k+1)}}{4} < x_2 \leq e^\sigma,$$

where the classical criterion ($x_2 > e^\sigma$) is silent, and which for every $k \geq 2$ strictly contains the determinantal window. (Above e^σ the classical argument applies; in fact no transcendental function of exact type (1) can have radius of convergence exceeding e^σ , so the window is also the natural habitat of such functions.)

For linear forms in 1 and $\zeta(5)$ one expects $\sigma = 5$. A hypothetical pair (A, V) annihilated by an irreducible operator of order k , with $\zeta(5)A - V$ holomorphic on $\mathbb{C} \setminus [x_2, \infty)$, proves the irrationality of $\zeta(5)$ provided $4x_2 > e^{5(k+2)/(k+1)}$:

k	2	3	4	6	10	∞
$x_2 >$	197	130	101	75	58	37.1

No construction approaching these specifications is currently known; the table is offered as a target. We note that mixed forms (involving $\zeta(3)$ as well) can be handled by the same method in the framework of linear independence statements, at the cost of enlarging the tuple.

5. EXPERIMENTAL OBSERVATIONS ON DENOMINATORS

The denominator term $\tau(\mathbf{b})$ of Theorem 2.1 is a worst-case bound. For the Apéry $\zeta(3)$ system (u_n, v_n the Apéry sequences, $\sigma = 3$, $k = 3$) we computed, in exact arithmetic, the kernel of the vanishing-to-order- n constraints for the tuples $\{1, V, \theta V\}$ and $\{1, V, \theta V, \theta^2 V\}$ (rational surrogates with the same types and the same derivative-tower correlation structure as the conditional P), for polynomial degrees up to 22 and vanishing orders $n^* \leq 67$, and measured the minimal denominator of the leading coefficient over the kernel. The observed rates are

$$\frac{\log \text{den}(\beta)}{n^*} \approx 1.76 \ (m = 3), \quad 1.89 \ (m = 4),$$

against the theoretical caps $\tau(\mathbf{b}) \cdot \frac{3\psi(n^*)}{\sigma n^*} \approx 2.5\text{--}2.6$ at these ranges; moreover the savings are concentrated at small primes ($p \leq n^*/4$), the primes in $(n^*/2, n^*]$ appearing with their full exponent 3. This is consistent with the supercongruence phenomena known for Apéry numbers and suggests that, for specific systems, provable “fine” denominator types substantially below (1) may be available, in the spirit anticipated in [2, Remarks 6.6.15 and 10.2.3]; any such improvement lowers the thresholds of Theorem 1.1 accordingly.

PROVENANCE AND DISCLOSURE

This note records the outcome of an extended exploration conducted as a human–AI collaboration. The direction of inquiry, the decisions at every fork, and the responsibility for circulating this note rest with the human correspondent (who, out of caution, leaves the byline empty for now); the mathematical drafting, the literature tracing, and the numerical experiments of §5 were carried out by the AI system Claude (Anthropic). Every statement and proof above is elementary modulo [2] and has been written so as to be checkable by hand against that paper. All of the machinery belongs, of course, to the authors of [2] and [4]; the present note merely turns a key that their work cut. Comments, corrections, and references

to prior occurrences of Theorem 1.1 are very welcome and will be gratefully incorporated or, if warranted, the note withdrawn.

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